

Connector from gate drivers

Includes multiple redundant SSO (six-switch-off) pathways:

- via gate drive reset
- via gate drive power enable 1
- via gate drive power enable 2

(Gate driver power enable implements 1002, with monitoring feedback).

feedback pins are 0-3.3v binary (3.3v expected when +12v present)

Connector to IO board

Digital Outputs provided here to IOBoard.

SSW-108-01-L-D

J7

SSW-108-01-L-D

J7

SSW-108-01-L-D

J7

SSW-108-01-L-D

J7

SSW-108-01-L-D

J7

SSW-108-01-L-D

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SSW-108-01-L-D

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J7

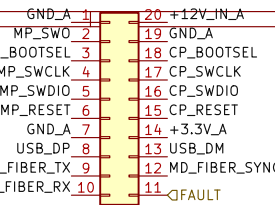
SSW-108-01-L-D

J7

SSW-108-01-L-D

J7

SSW-108-01-L-D



This design has been implemented to provide multiple redundant pathways to SSO (Six-switch off) / STO (Safe torque-off). SSO/STO is the default safe-state of the inverter please (see Hazard And Risk Assessment documentation)

Additionally, while best effort has been made to implement functional safety best practices, please note that no formal independent review has taken place, and therefore no formal ASIL/SIL compliance is claimed.

Temperature Sensors



File: TemperatureSensors.kicad_sch

CONNECTOR FROM THIS GATE DRIVER

TO MAIN CONTROL BOARD, BELOW

Voltage Divider Network3

GND_A

File: VDivNetHalf.kicad_sch

Voltage Divider Network2

GND_A

File: VDivNet.kicad_sch

GND_A

File: VDivNet.kicad_sch

GND_A

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GND_A

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GND_A

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TO MAIN CONTROL BOARD, BELOW

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GND_A

File: VDivNetHalf.kicad_sch

Voltage Divider Network2

GND_A

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GND_A

File: VDivNet.kicad_sch

GND_A

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Voltage Divider Network2

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GND_A

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GND_A

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GND_A

File: VDivNet.kicad_sch

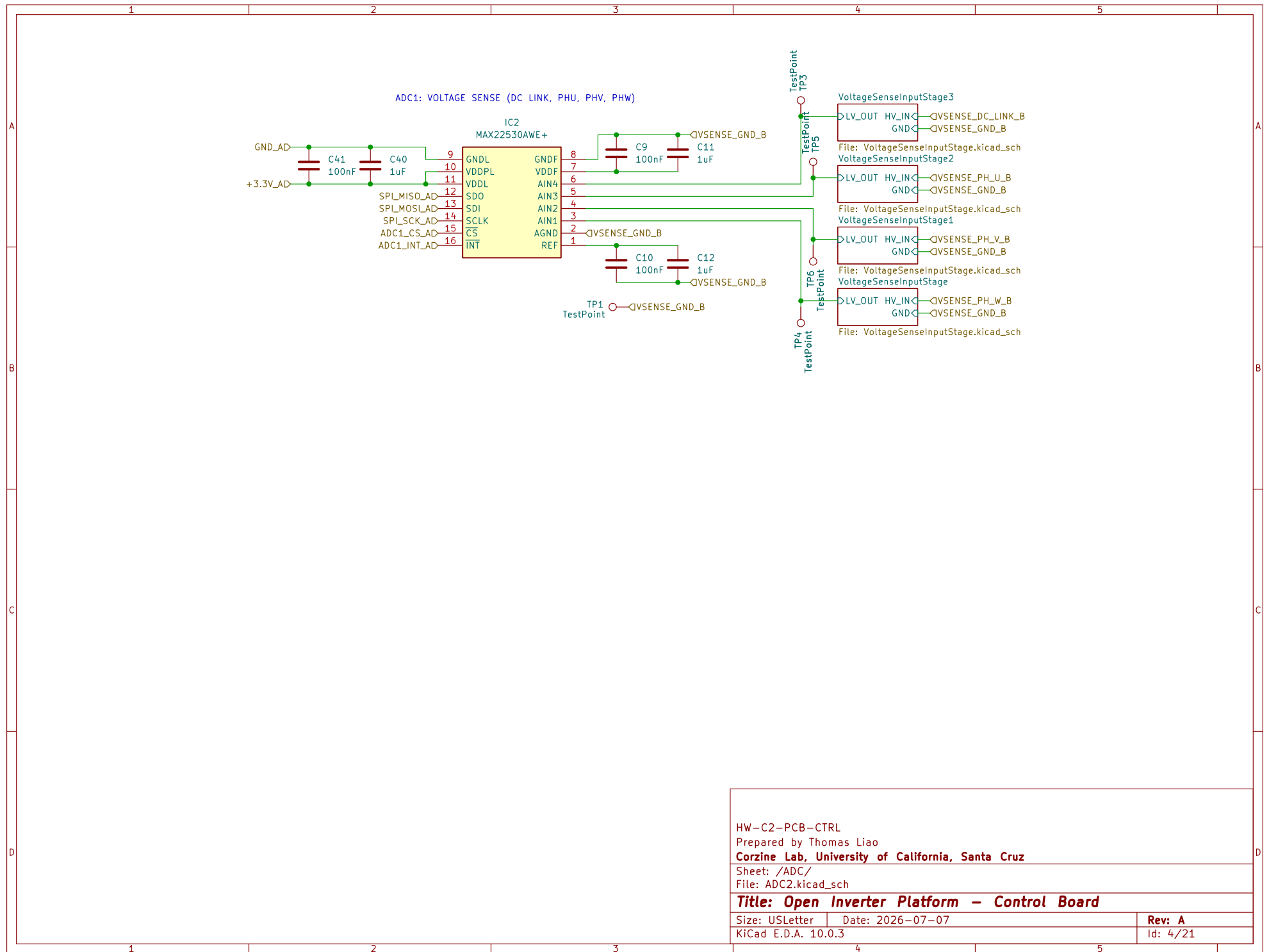
GND_A

File: VDivNet.kicad_sch

GND_A

File: VDivNet.kicad_sch

GND_A



HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/

File: ADC2.kicad_sch

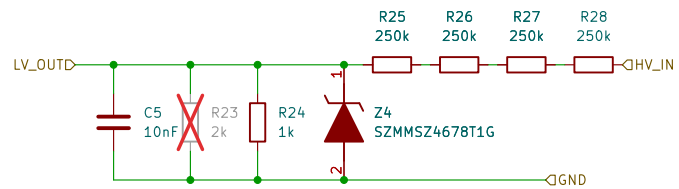
Title: Open Inverter Platform – Control Board

Size: USLetter Date: 2026-07-07

KiCad E.D.A. 10.0.3

Rev: A

Id: 4/21



ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/VoltageSenseInputStage/

File: VoltageSenseInputStage.kicad_sch

Title: Open Inverter Platform – Control Board

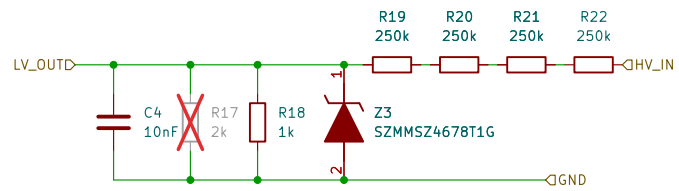
Size: USLetter

Date: 2026-07-07

Rev: A

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Id: 2/21



ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/VoltageSenseInputStage1/

File: VoltageSenseInputStage.kicad_sch

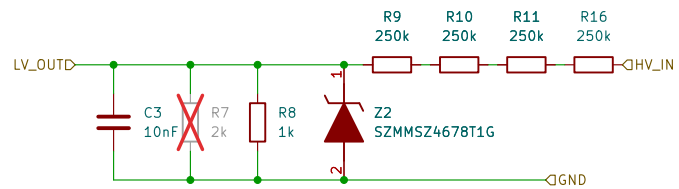
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Size: USLetter Date: 2026-07-07

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Rev: A

Id: 3/21



ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/VoltageSenseInputStage2/

File: VoltageSenseInputStage.kicad_sch

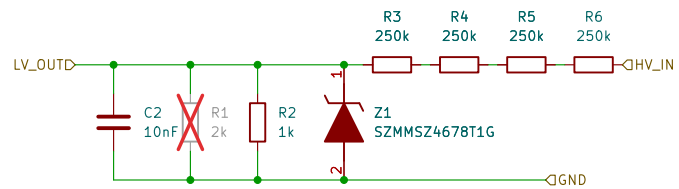
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Size: USLetter Date: 2026-07-07

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Rev: A

Id: 5/21



ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/VoltageSenseInputStage3/

File: VoltageSenseInputStage.kicad_sch

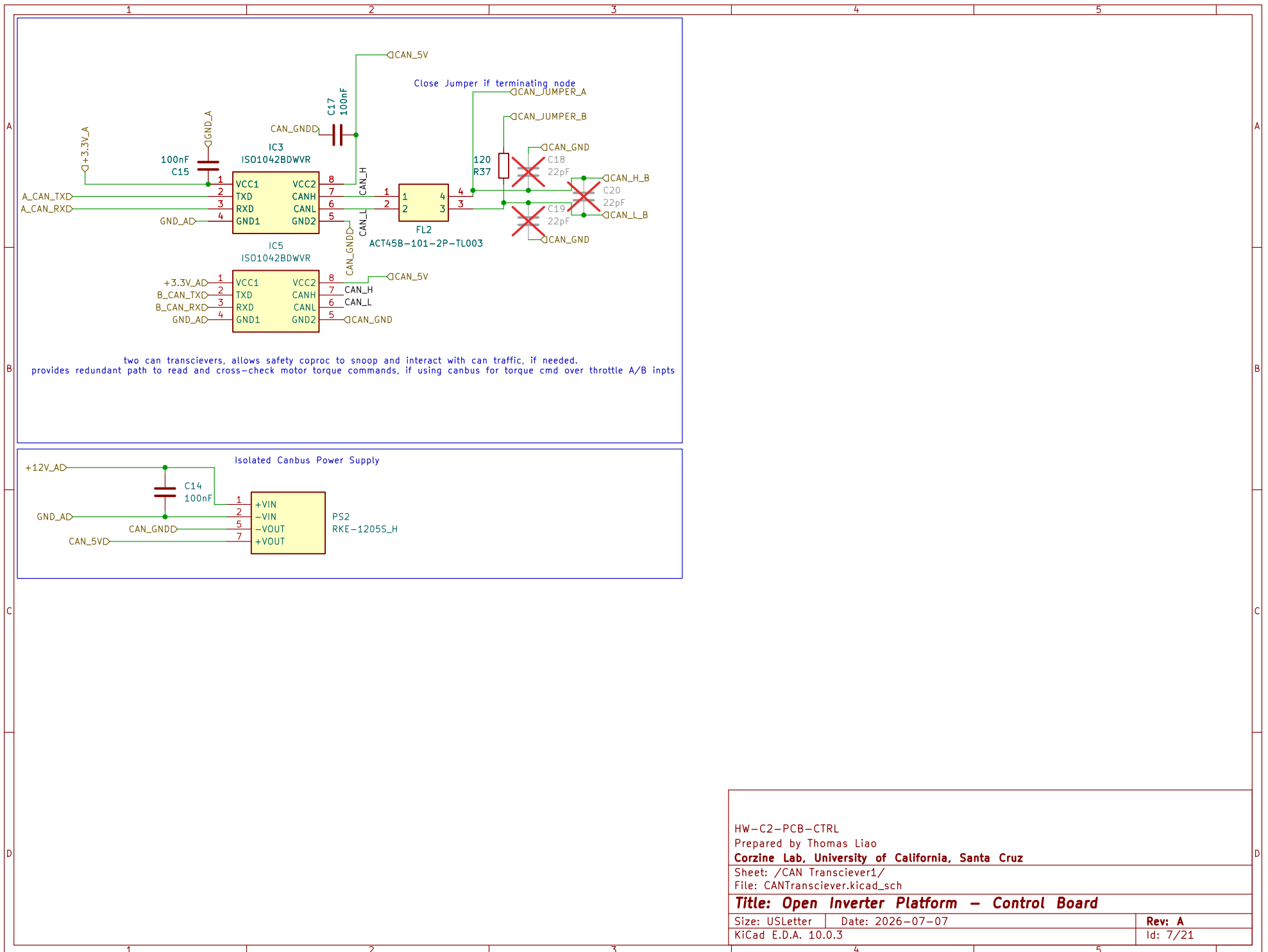
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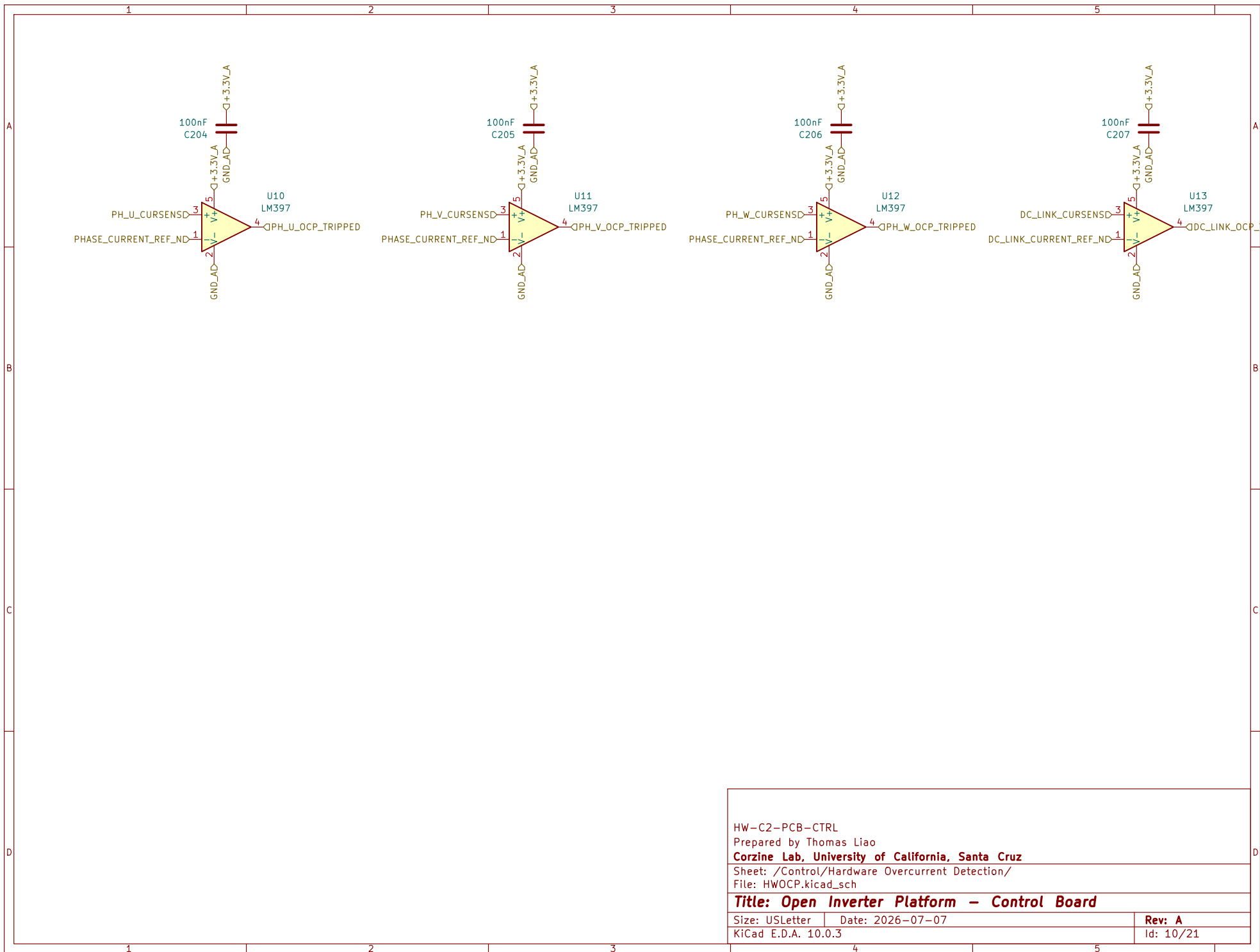
Size: USLetter Date: 2026-07-07

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Rev: A

Id: 6/21





HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /Control/Hardware Overcurrent Detection/

File: HWOCP.kicad_sch

Title: Open Inverter Platform – Control Board

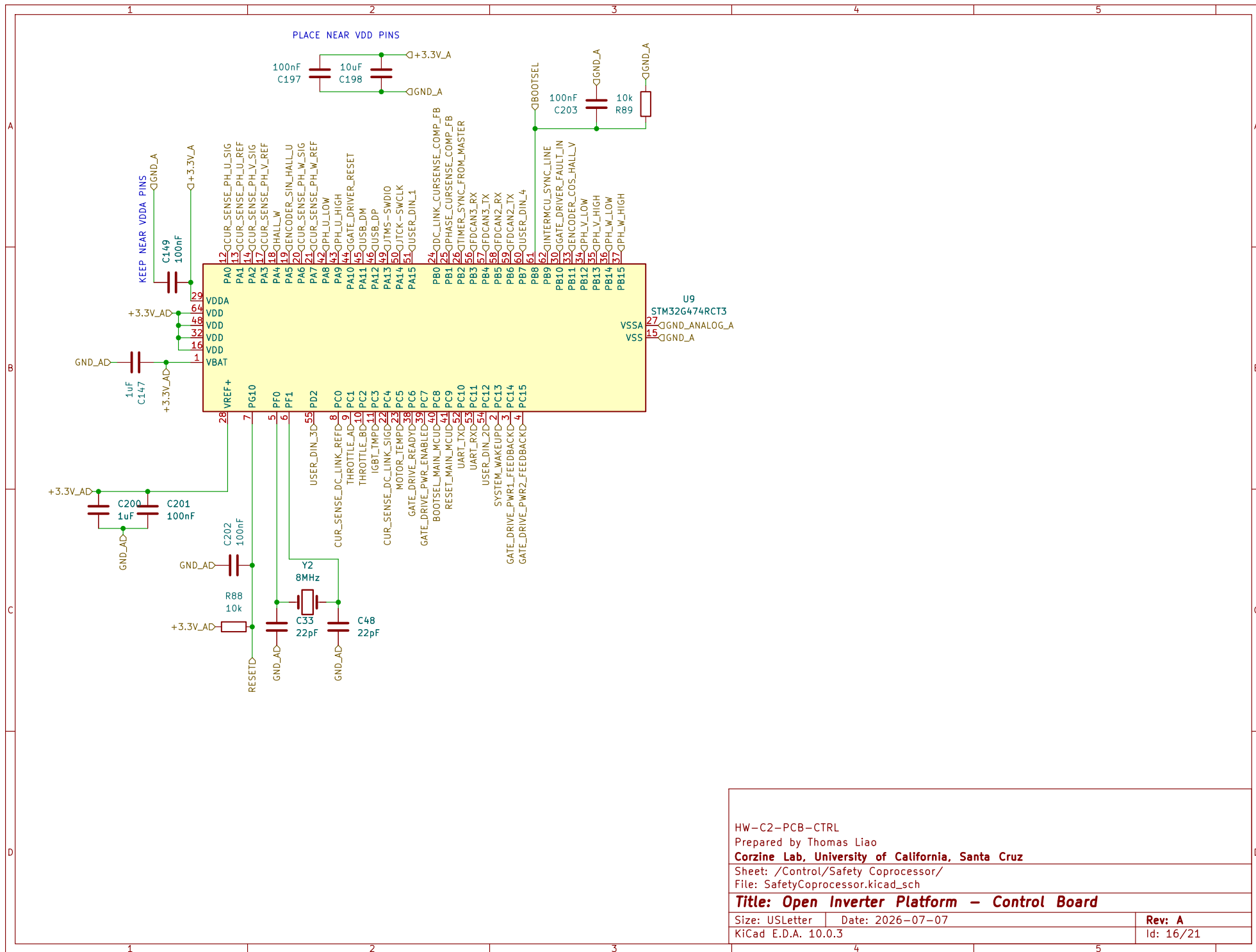
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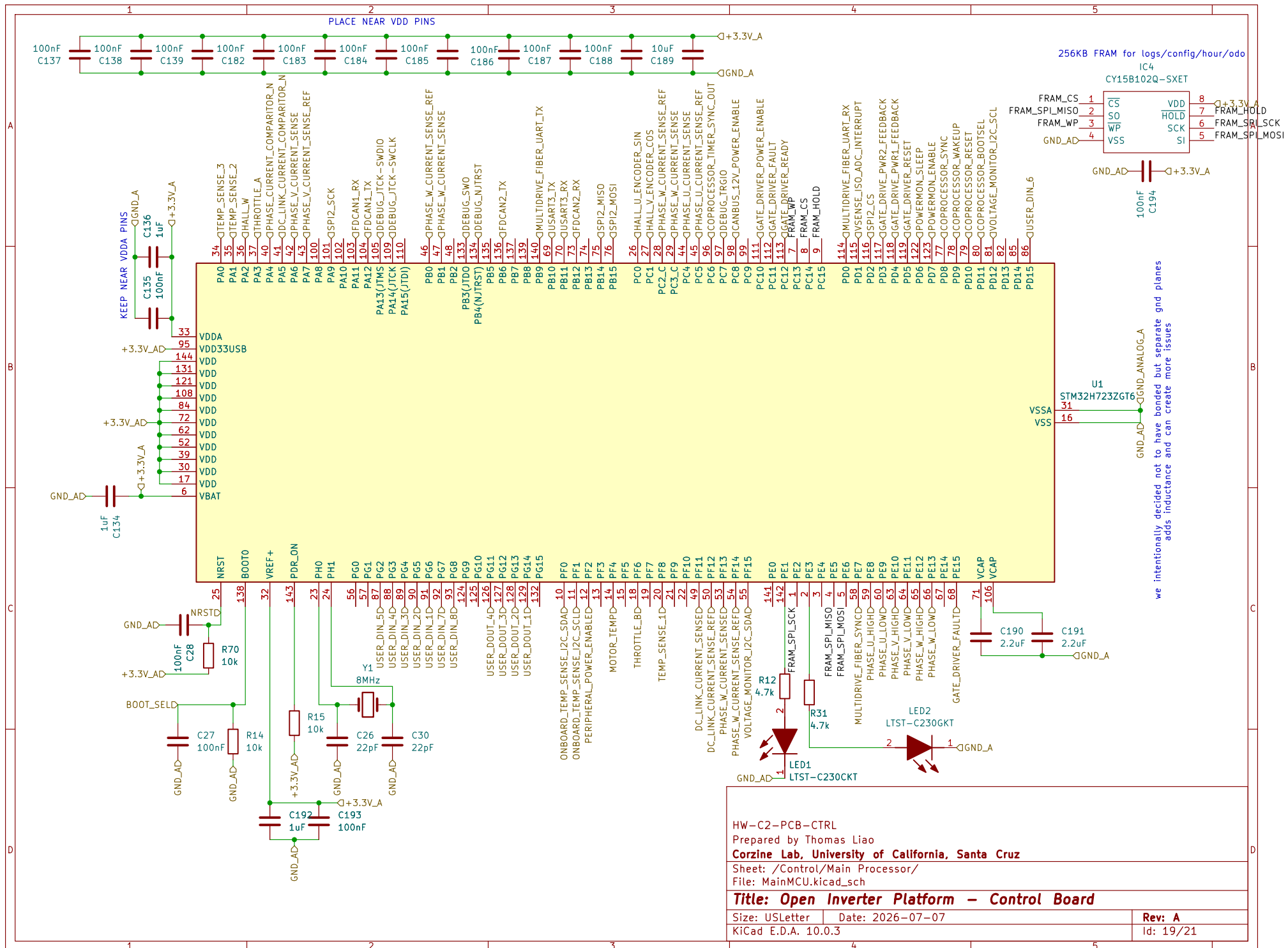
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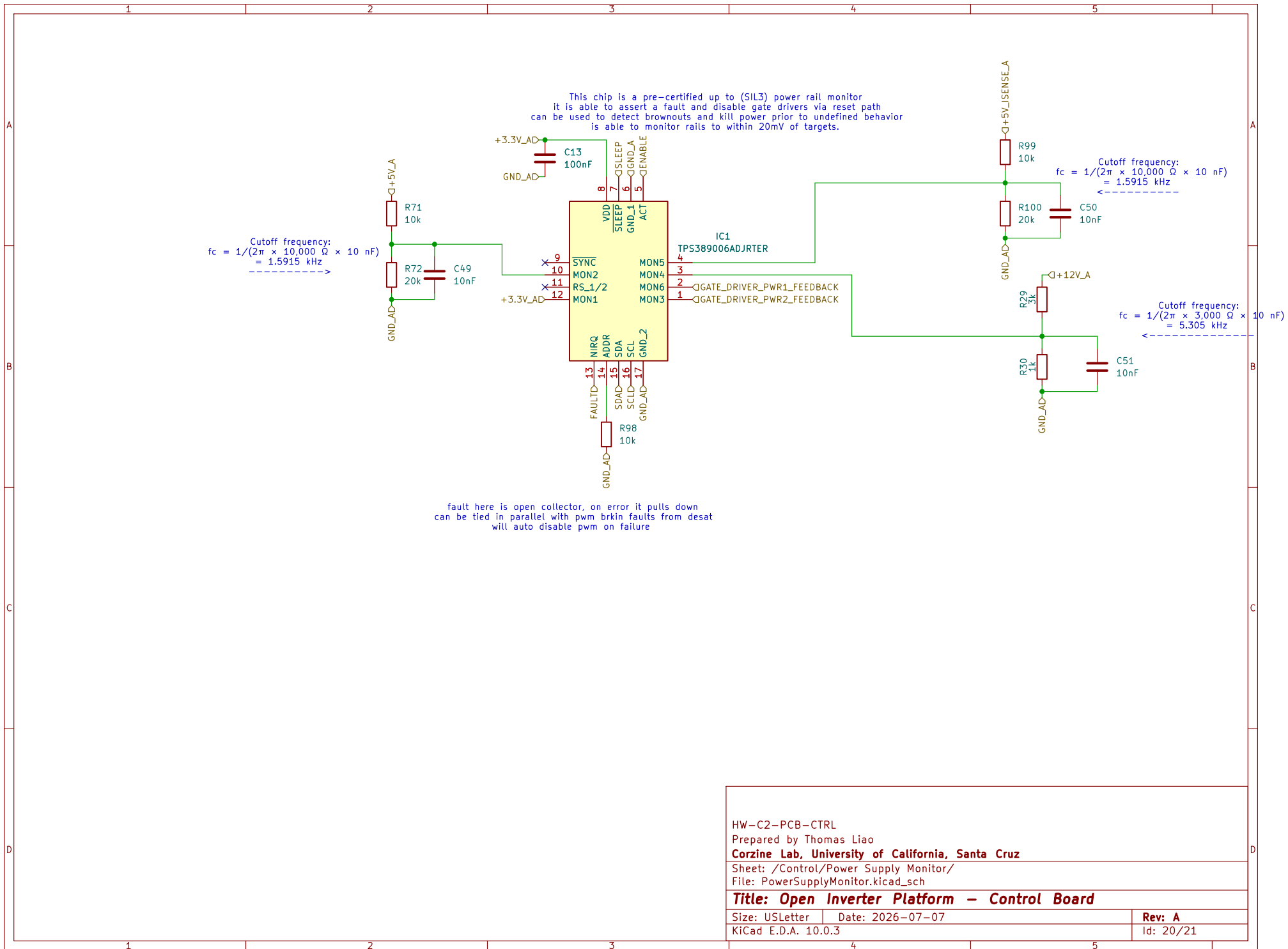
Rev: A

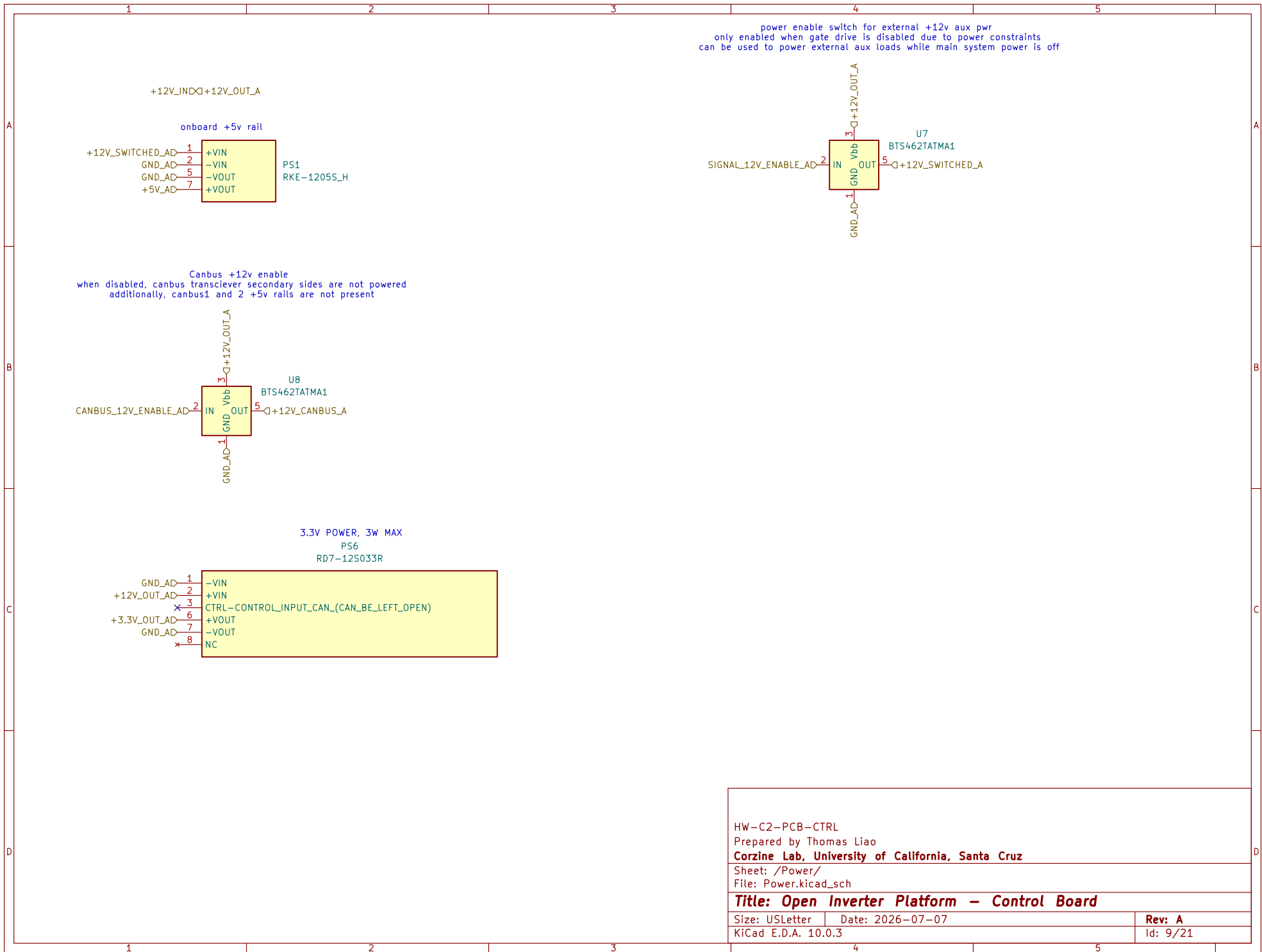
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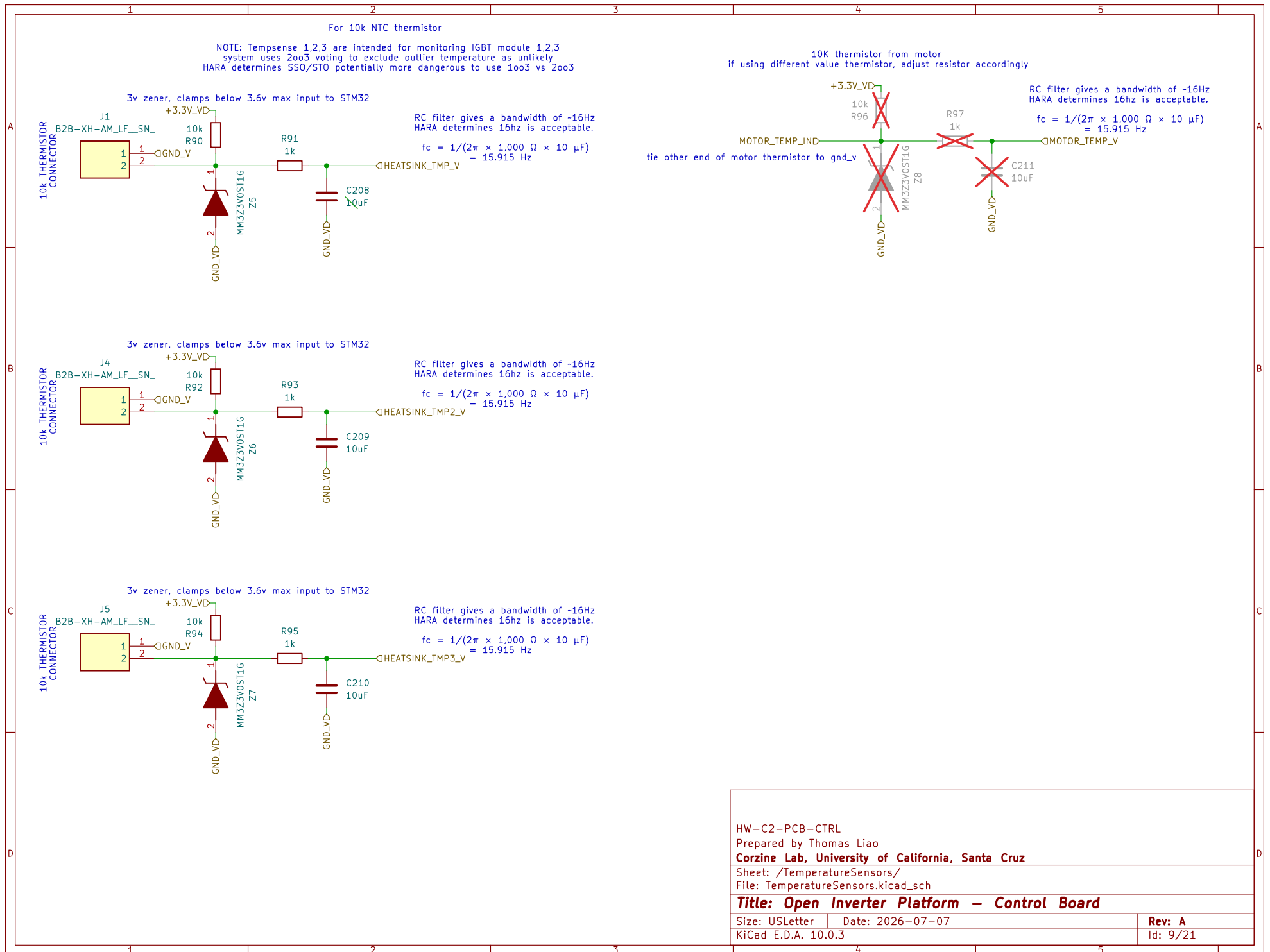
Id: 10/21











HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /TemperatureSensors/

File: TemperatureSensors.kicad_sch

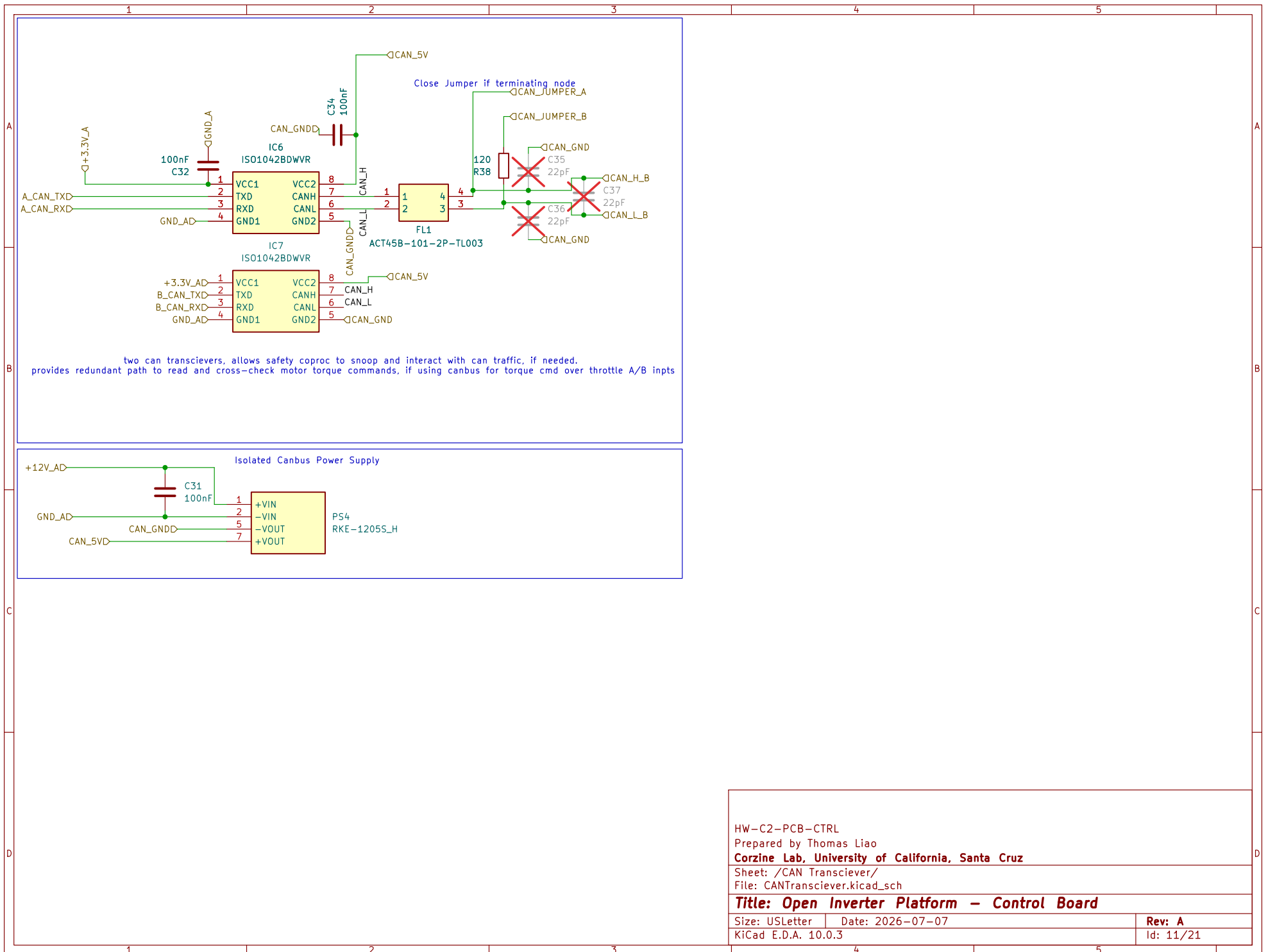
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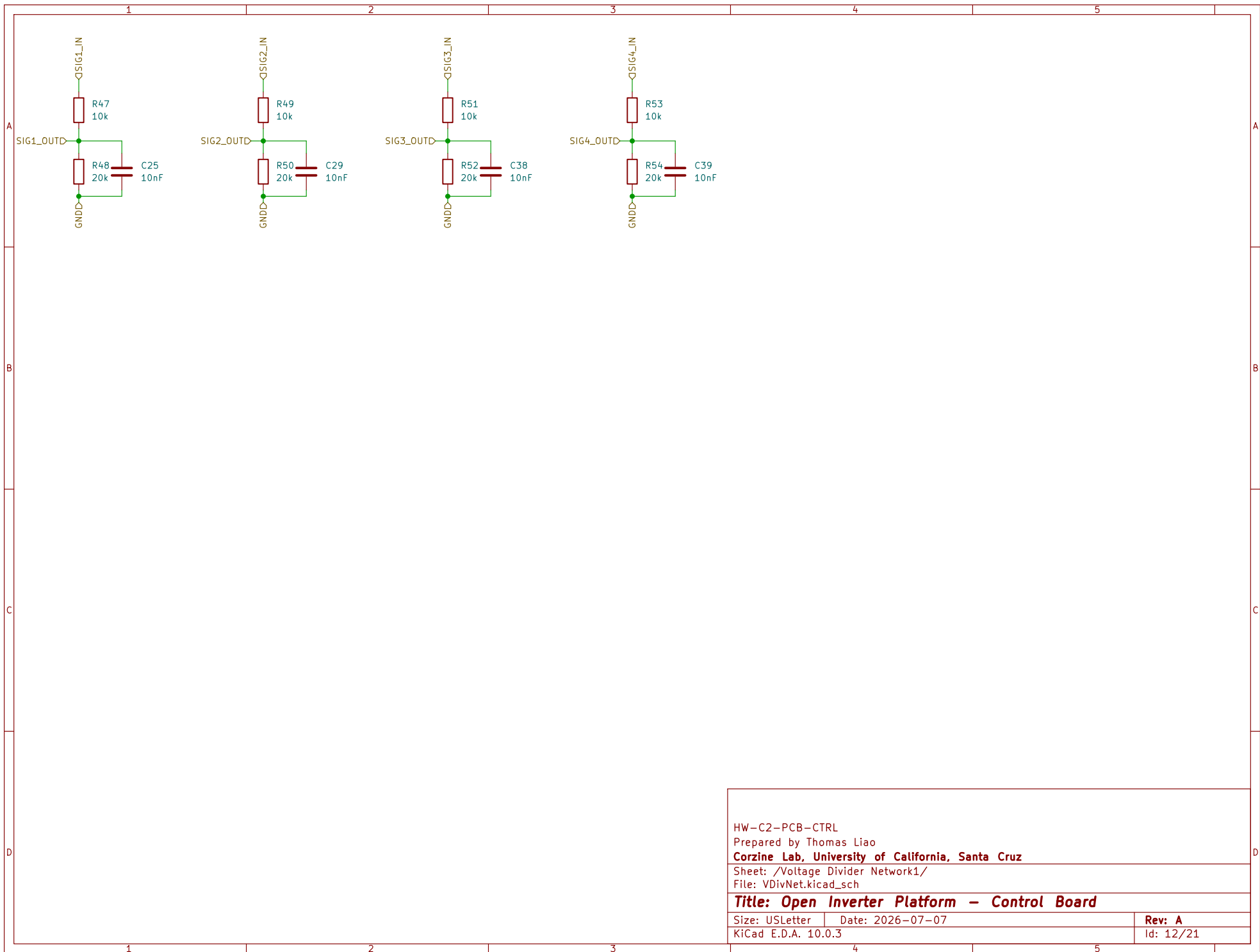
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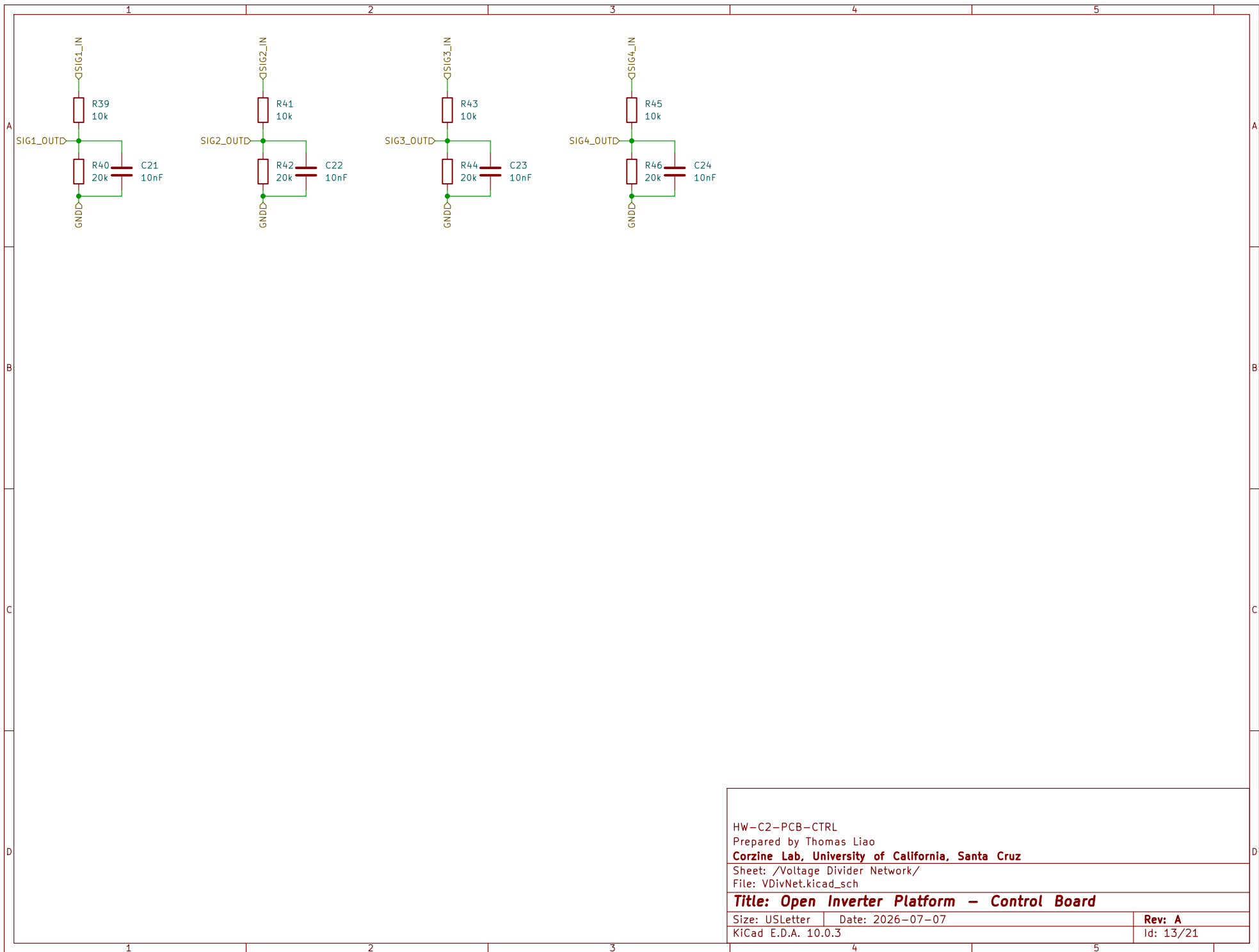
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Rev: A

Id: 9/21







HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /Voltage Divider Network/

File: VDivNet.kicad_sch

Title: Open Inverter Platform – Control Board

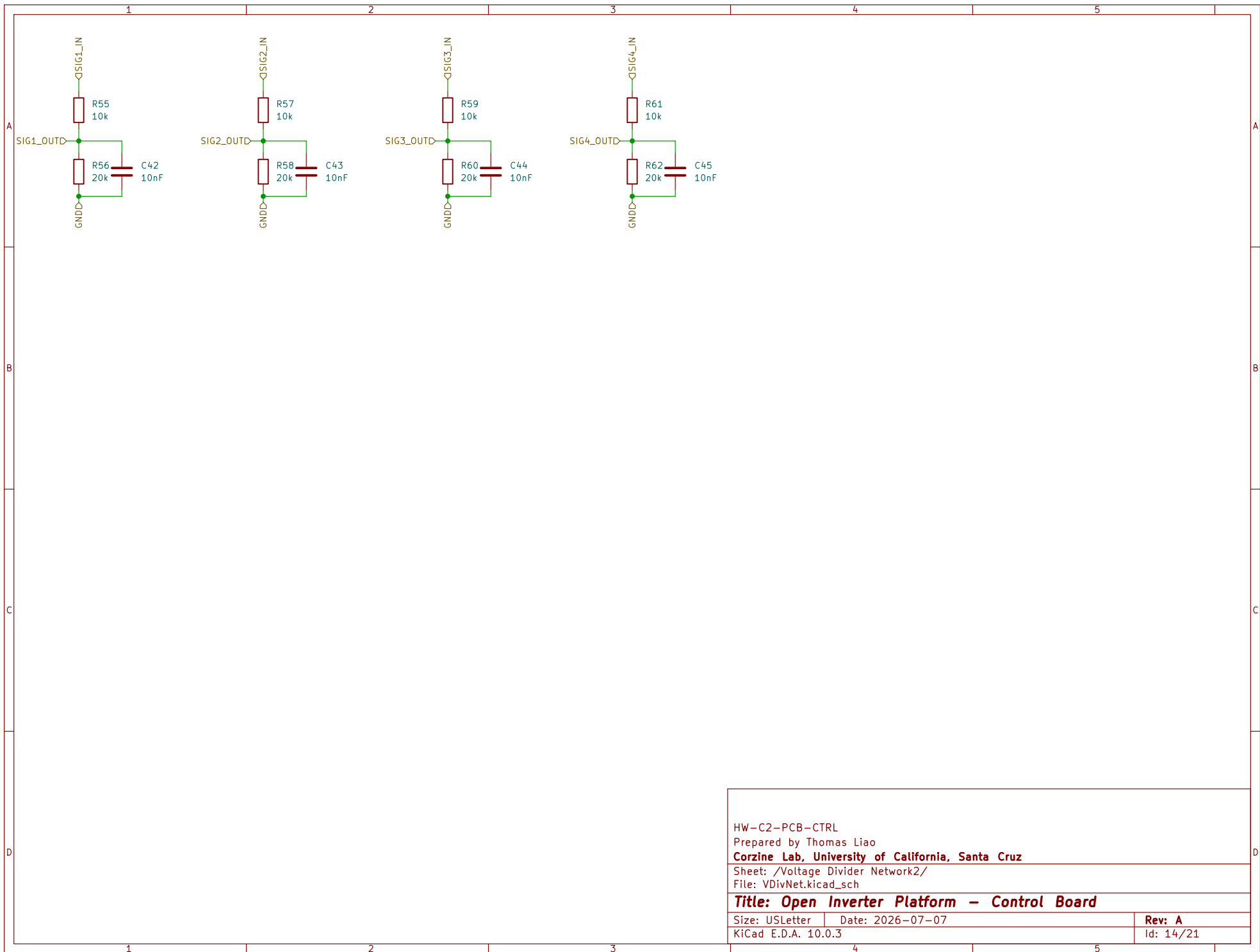
Size: USLetter

Date: 2026-07-07

Rev: A

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Id: 13/21



HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /Voltage Divider Network2/

File: VDivNet.kicad_sch

Title: Open Inverter Platform – Control Board

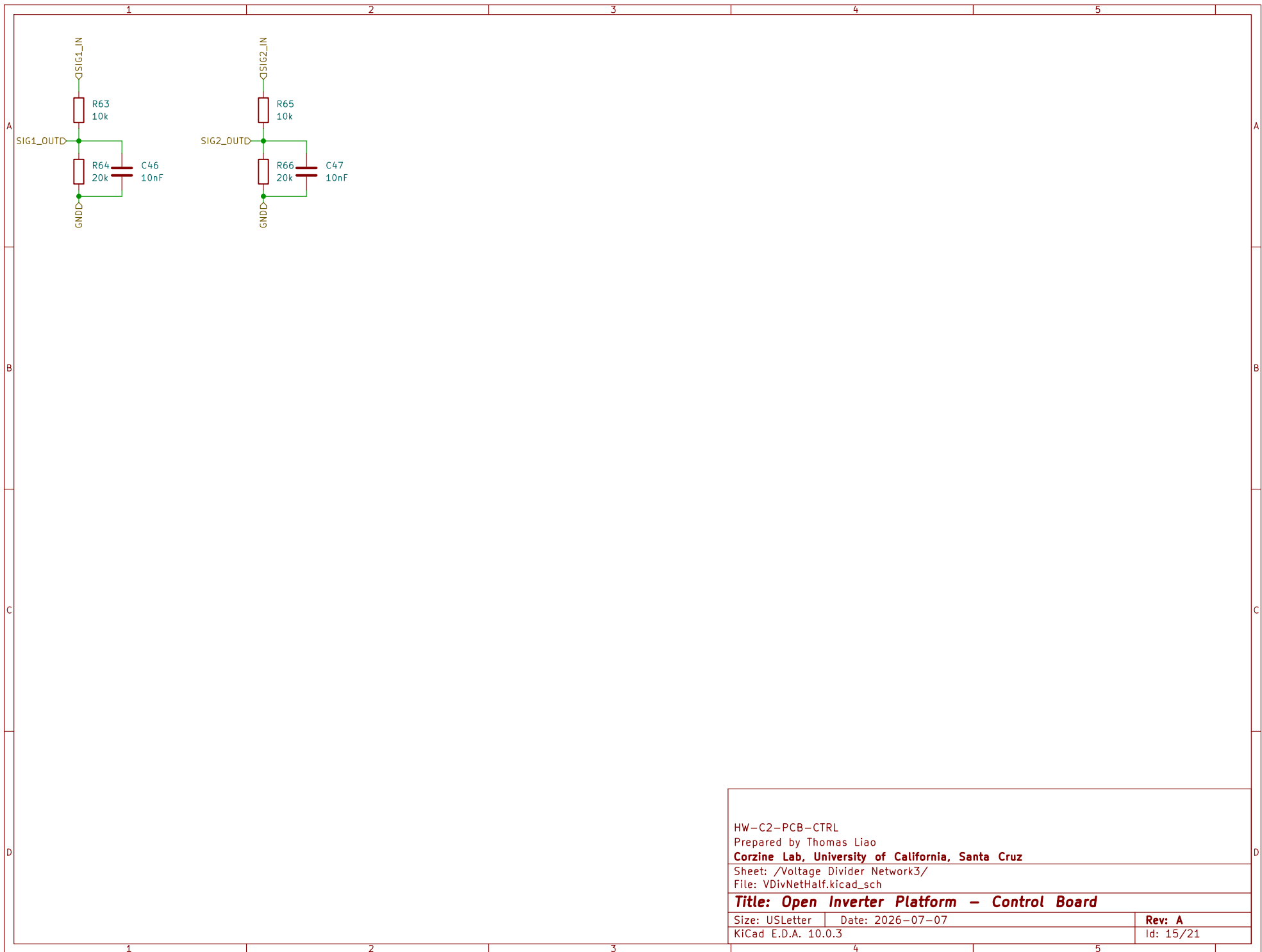
Size: USLetter

Date: 2026-07-07

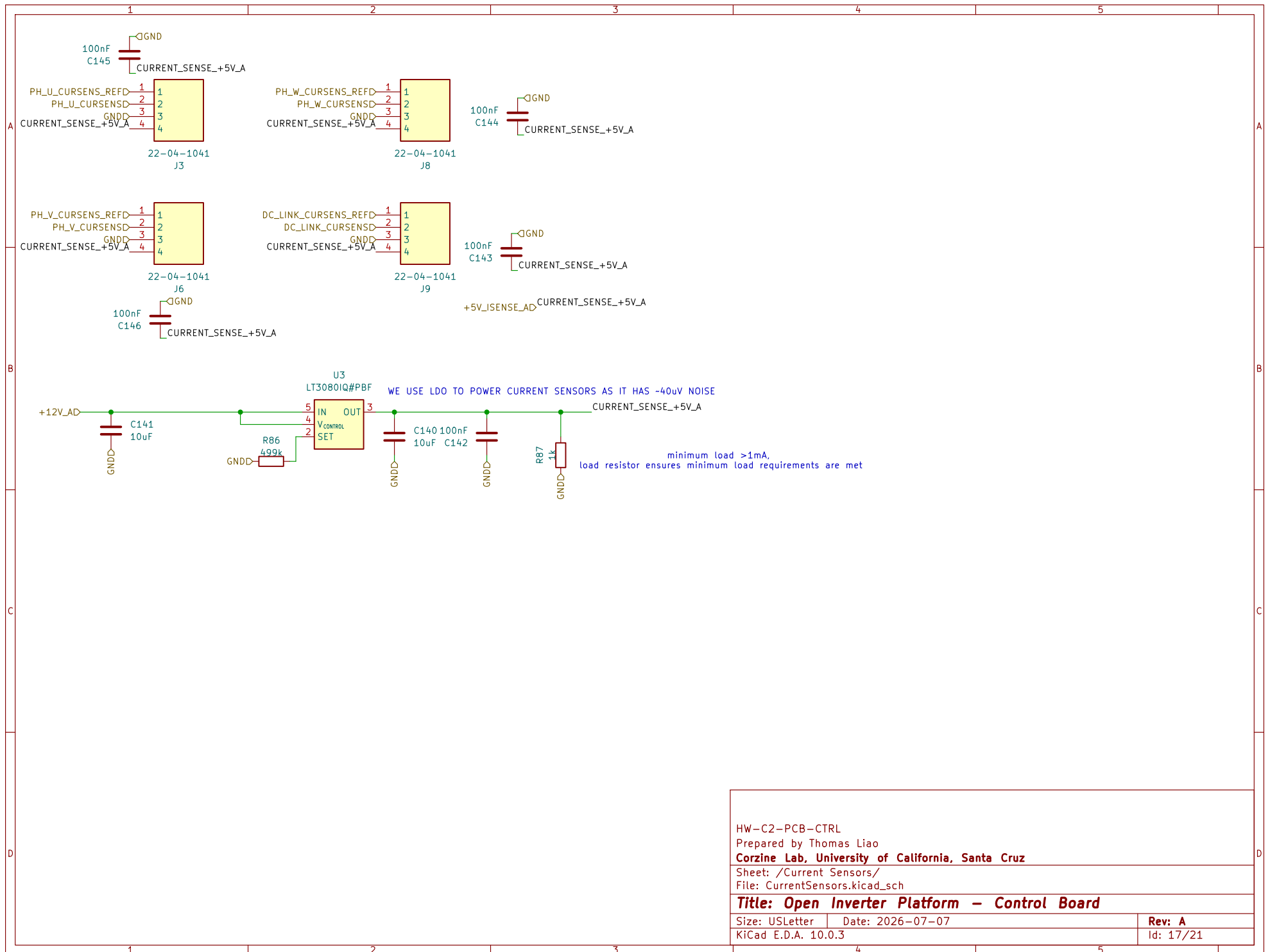
Rev: A

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Id: 14/21



HW-C2-PCB-CTRL		
Prepared by Thomas Liao		
Corzine Lab, University of California, Santa Cruz		
Sheet: /Voltage Divider Network3/		
File: VDivNetHalf.kicad_sch		
Title: Open Inverter Platform – Control Board		
Size: USLetter	Date: 2026-07-07	Rev: A
KiCad E.D.A. 10.0.3	Id: 15/21	



HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /Current Sensors/

File: CurrentSensors.kicad_sch

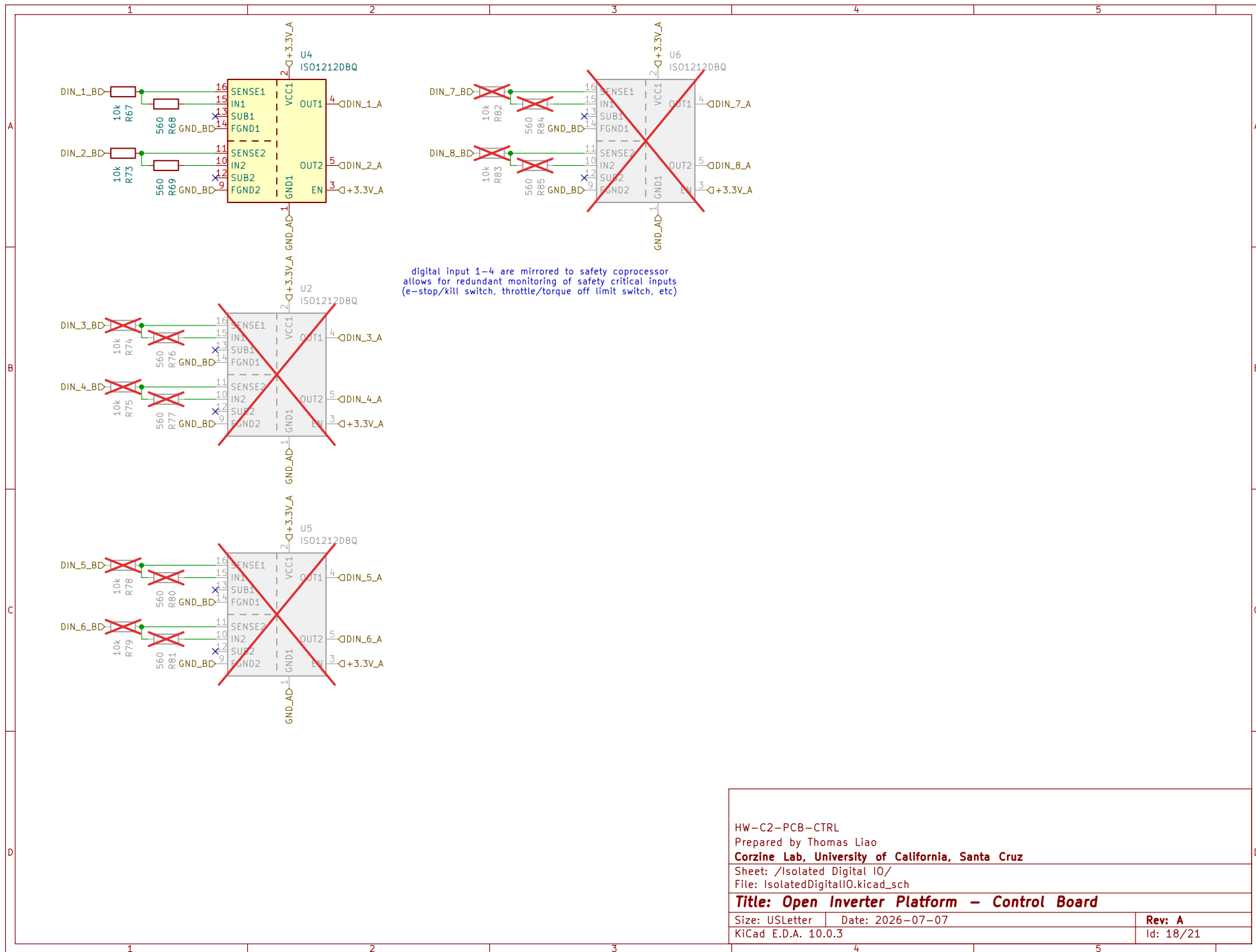
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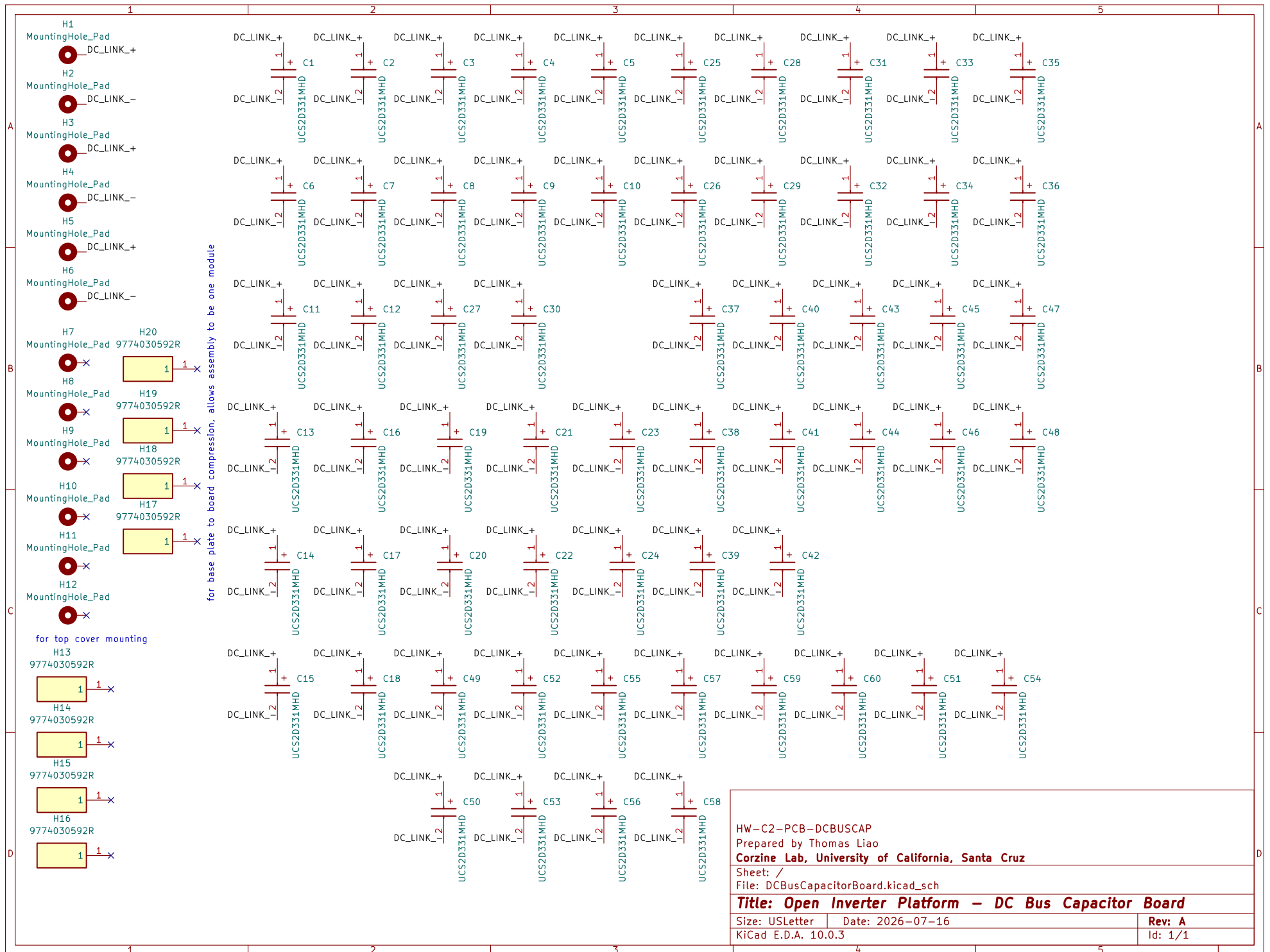
Size: USLetter Date: 2026-07-07

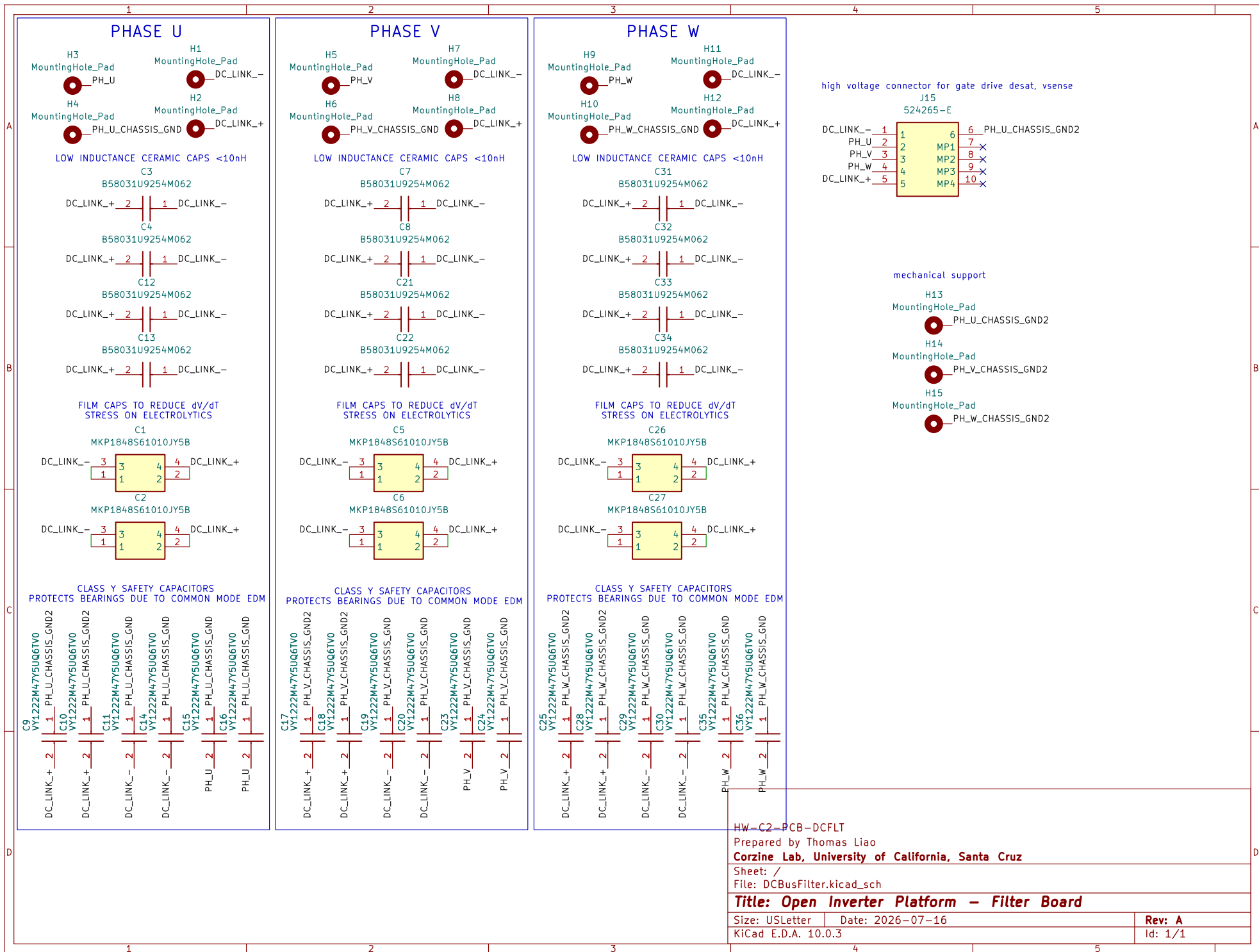
KiCad E.D.A. 10.0.3

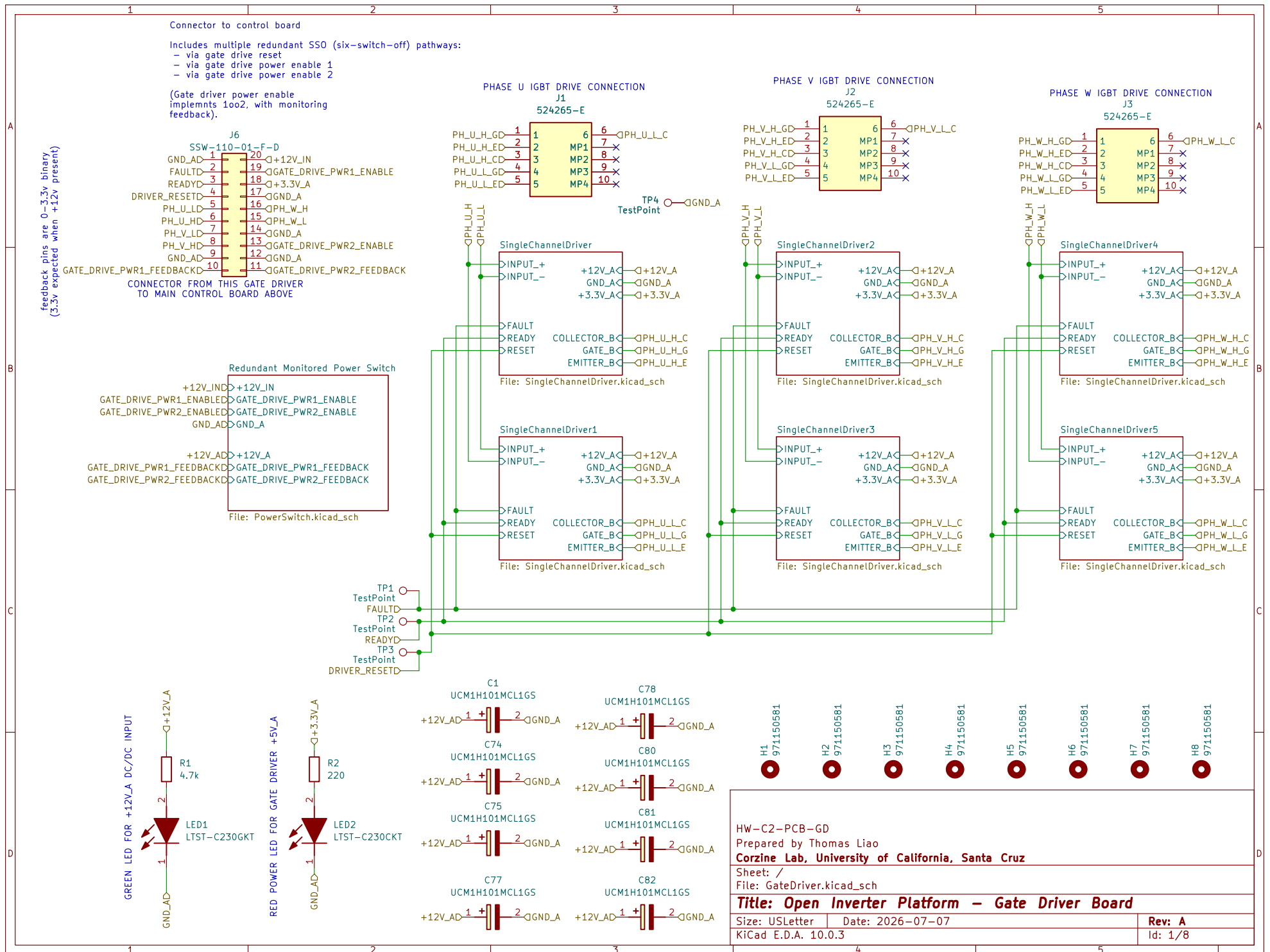
Rev: A

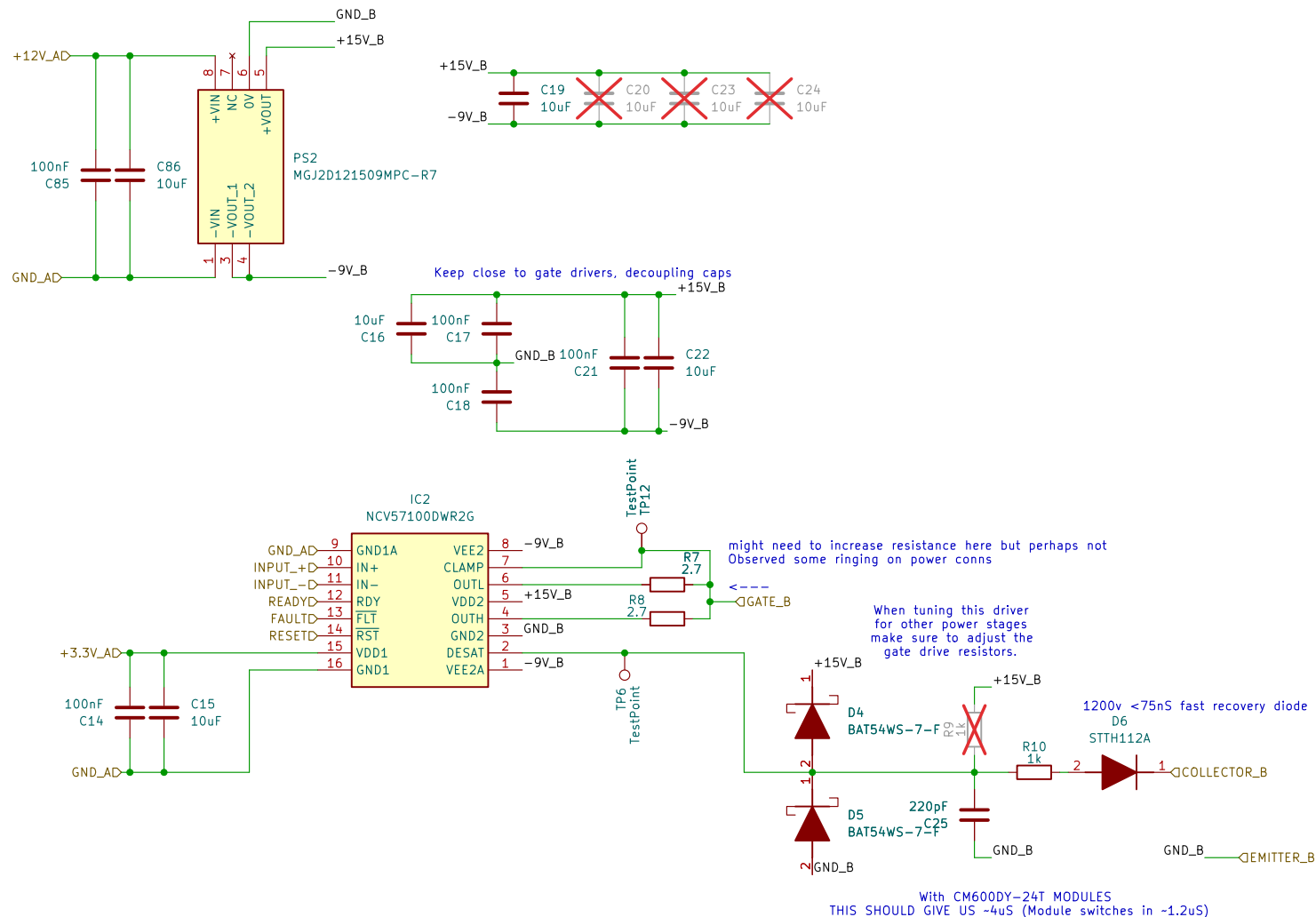
Id: 17/21











HW-C2-PCB-GD

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /SingleChannelDriver2/

File: SingleChannelDriver.kicad_sch

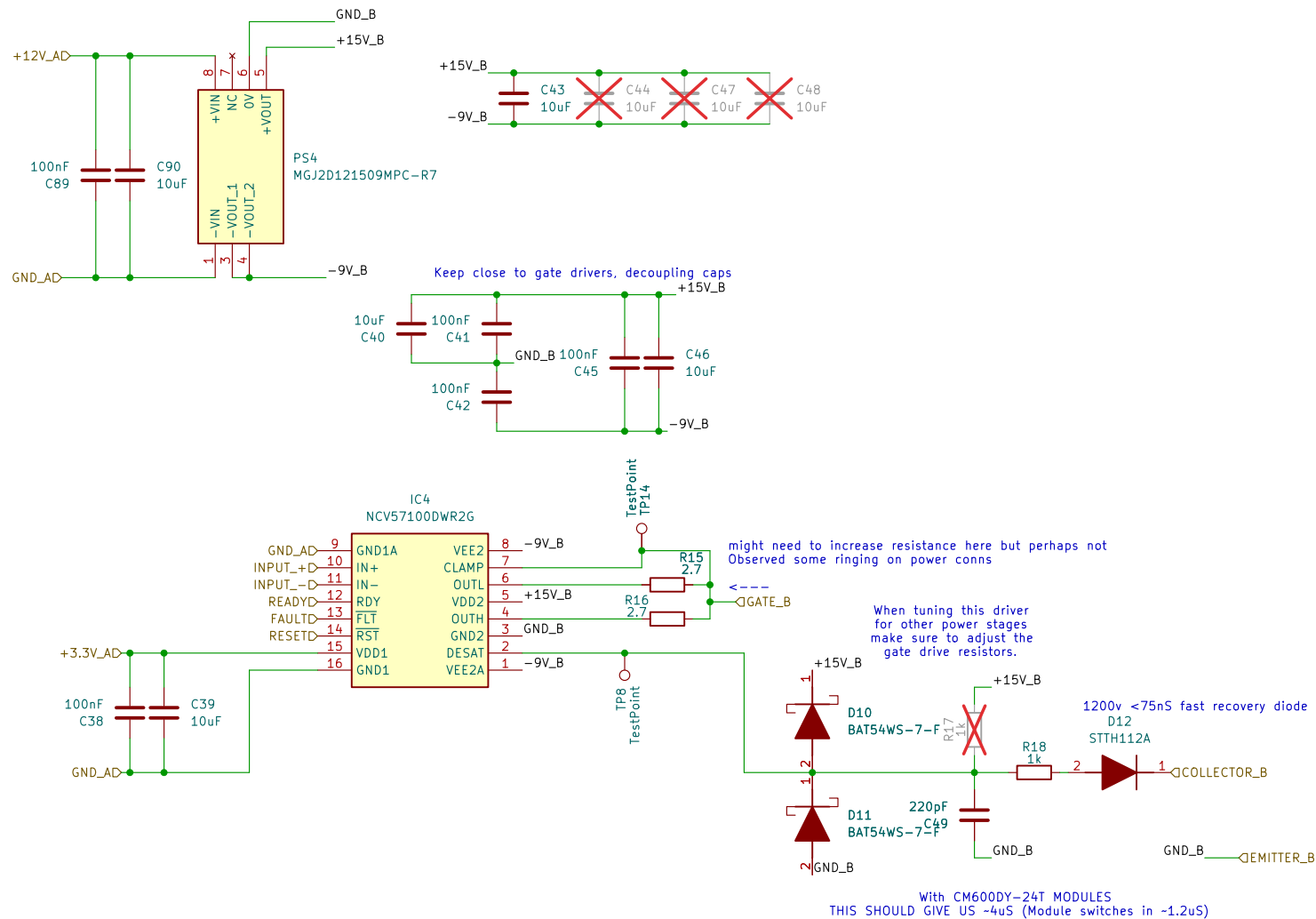
Title: Open Inverter Platform - Gate Driver Board

Size: USLetter Date: 2026-07-07

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Rev: A

Id: 3/8



HW-C2-PCB-GD

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /SingleChannelDriver3/

File: SingleChannelDriver.kicad_sch

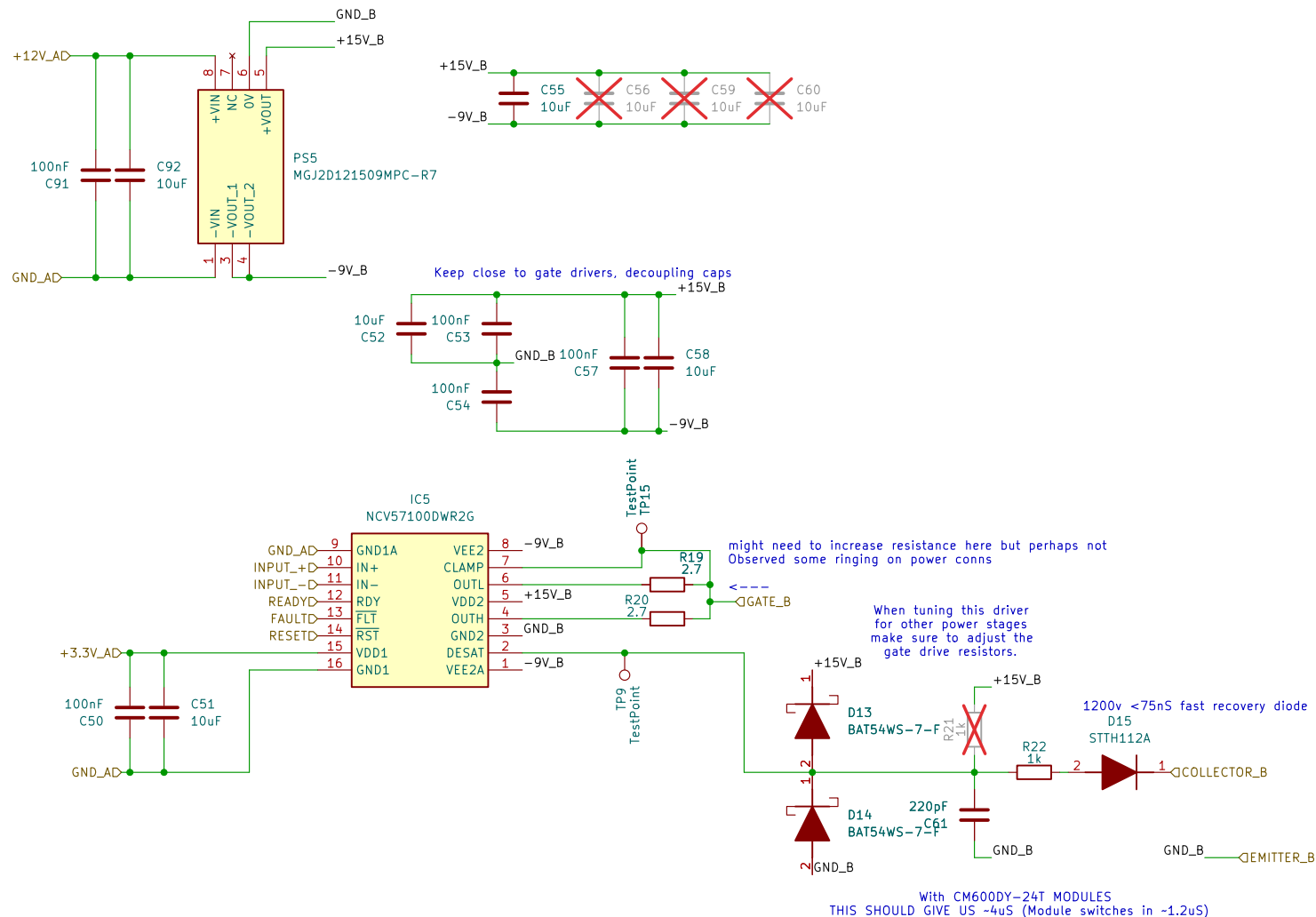
Title: Open Inverter Platform - Gate Driver Board

Size: USLetter Date: 2026-07-07

KiCad E.D.A. 10.0.3

Rev: A

Id: 5/8



HW-C2-PCB-GD

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /SingleChannelDriver4/

File: SingleChannelDriver.kicad_sch

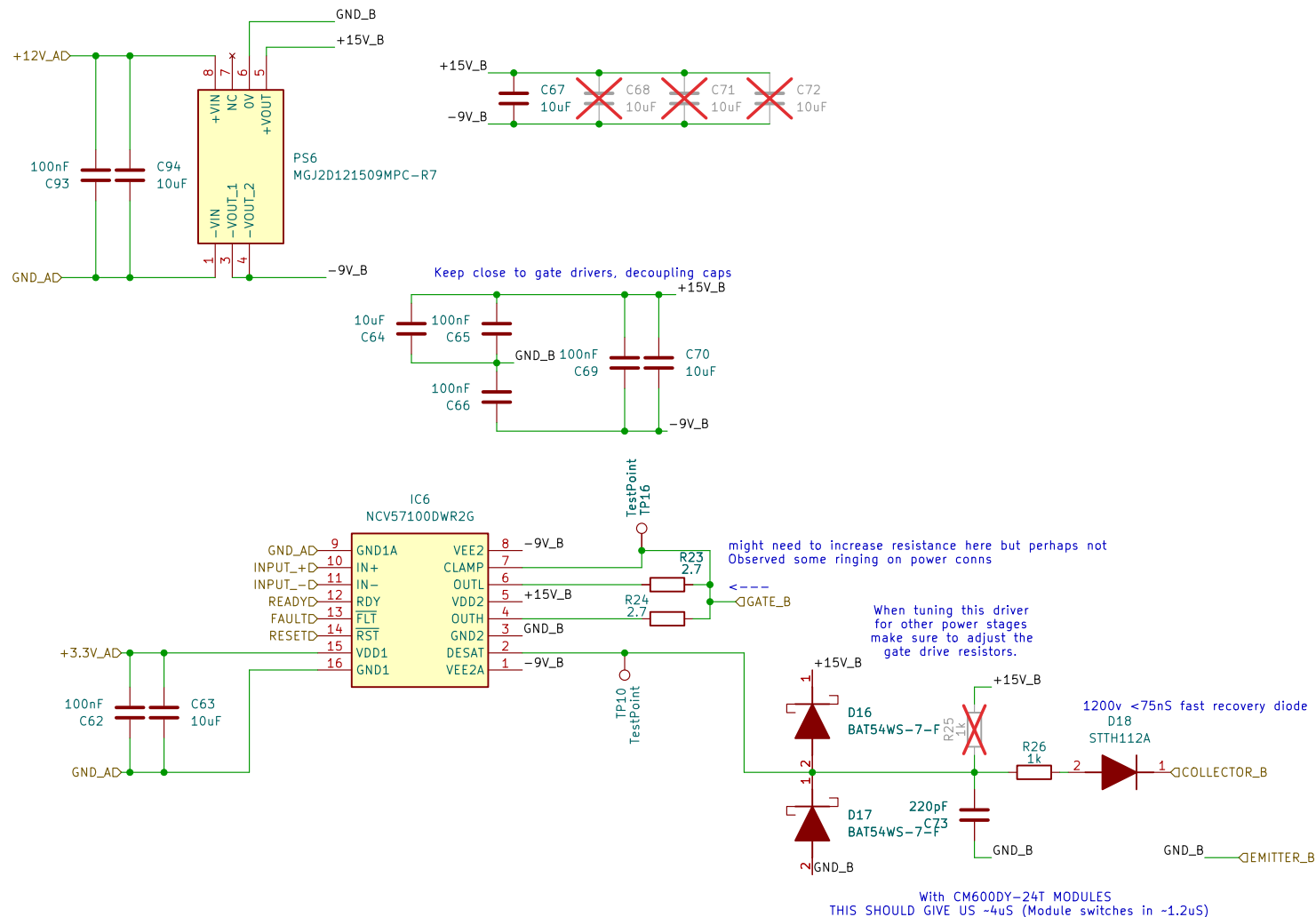
Title: Open Inverter Platform - Gate Driver Board

Size: USLetter Date: 2026-07-07

KiCad E.D.A. 10.0.3

Rev: A

Id: 6/8



HW-C2-PCB-GD

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /SingleChannelDriver5/

File: SingleChannelDriver.kicad_sch

Title: Open Inverter Platform - Gate Driver Board

Size: USLetter Date: 2026-07-07

KiCad E.D.A. 10.0.3

Rev: A

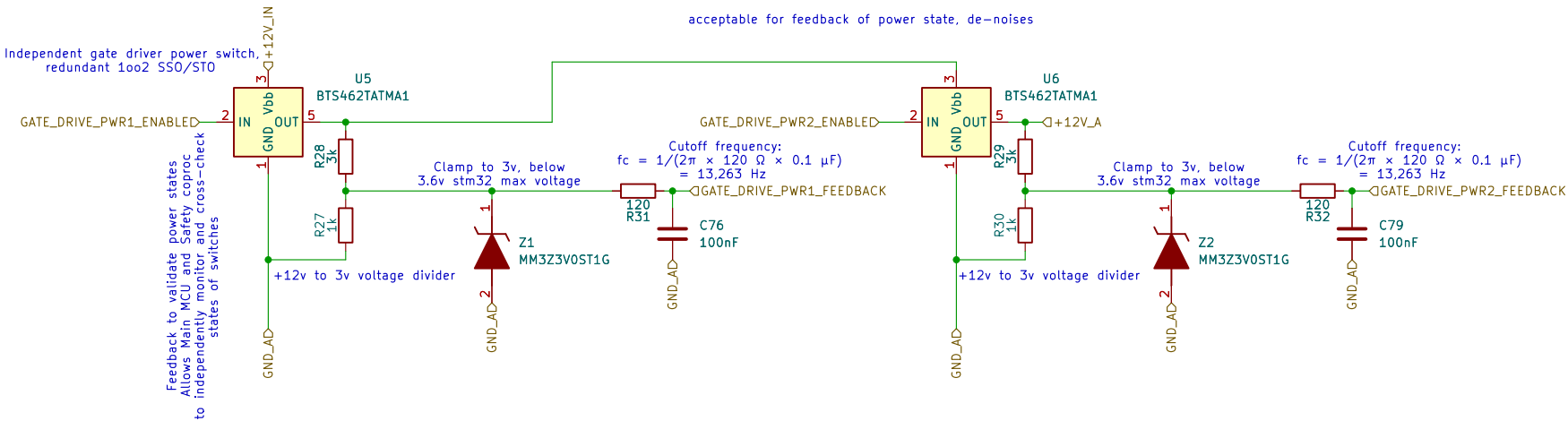
Id: 7/8

This circuit implements a redundant pathway for both mcus to implement Six Switch Open / Safe Torque Off in addition to regular pwm break and gate driver reset.

The following switches are independent systems with feedback monitoring. Each MCU cross monitors returned inputs, and can validate if the switch failed to respond as commanded – thus removing latent failures from this safety subsystem.

cutoff period (1/freq) ->
75.4 uS

acceptable for feedback of power state, de-noises



HW-C2-PCB-GD

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /Redundant Monitored Power Switch/

File: PowerSwitch.kicad_sch

Title: Open Inverter Platform – Gate Driver Board

Size: USLetter

Date: 2026-07-07

Rev: A

KiCad E.D.A. 10.0.3

Id: 8/8

